

AYUSH KUMAR

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Professional Summary

Electronics and Communication Engineering student specializing in embedded systems design, machine learning applications, and IoT solutions. Experienced in developing production-grade systems using Python, C/C++, and microcontroller architectures (Arduino, 8051). Built end-to-end ML pipeline achieving 89% accuracy for career recommendation system and designed real-time embedded applications with motor control and sensor integration. Competitive programmer with final round selection at IIT Madras Quantathon and active participation in national-level space technology hackathons. Strong foundation in circuit design, PWM control systems, and full-stack development with Flask.

Education

RV Institute of Technology and Management <i>Bachelor of Engineering in Electronics and Communication</i>	Aug 2024 – Aug 2028
IIT Ropar <i>Minor in AI</i>	Sep 2024 – Oct 2025
Trident Public School <i>Intermediate</i>	May 2023
Paramount Academy <i>Matriculate</i>	Aug 2021

Experience

Campus Ambassador <i>TRYST, IIT Delhi · Internship</i>	Feb 2026 – Mar 2026 <i>India · Remote</i>
- Represented TRYST, IIT Delhi's annual technical fest, driving awareness and student participation from RV Institute of Technology and Management. - Coordinated outreach campaigns, shared event updates, and encouraged peer registrations for competitions, workshops, and keynote sessions. - Demonstrated leadership and communication skills by bridging the organizing committee and the local student community effectively.	
Campus Ambassador <i>eDC (Entrepreneurship Development Cell), IIT Delhi · Full-time</i>	Dec 2025 – Feb 2026 <i>IIT Delhi · Remote</i>
- Promoted eDC IIT Delhi's flagship events including BECon 2026 through targeted social media campaigns and peer-to-peer outreach across campuses. - Boosted student engagement and registrations for entrepreneurship-focused programs, expanding eDC's reach beyond IIT Delhi. - Ranked among top-performing ambassadors for consistent outreach metrics, networking efforts, and event publicization. - Developed practical skills in digital marketing, entrepreneurship communication, and community engagement within a high-impact IIT ecosystem.	

Skills

Programming Languages: C, C++, Python, Embedded C
Web Technologies: Flask, HTML, CSS, Bootstrap
Machine Learning: Scikit-learn, Decision Trees, SVM, Random Forest, Gradient Boosting, Data Preprocessing
Embedded Systems: Arduino, 8051 Microcontroller, ATmega, Sensor Integration
Electronics & Hardware: Circuit Design, PWM Control, Motor Drivers (L293D), LCD Interfacing, IoT Systems
Simulation & Design Tools: Proteus, Git, GitHub
Core Competencies: Product Management, Cybersecurity, Ethical Hacking, Venture Capital, Data Analysis
Soft Skills: Problem Solving, Teamwork, Adaptability, Time Management, Effective Communication, Attention to Detail

Projects

Career X - Intelligent Education Recommendation System <i>Python, Flask, ML, Scikit-learn</i>
- Developed an intelligent career recommendation system using machine learning to provide personalized education and career guidance based on student academic performance and skills. - Implemented and compared multiple ML algorithms (Decision Trees, SVM, Random Forest, Gradient Boosting), achieving 89.1% accuracy with Gradient Boosting classifier. - Built a user-friendly Flask web application with HTML, CSS, and Bootstrap frontend, enabling students to receive top 3 career recommendations with confidence scores. - Performed comprehensive data preprocessing and feature engineering on Kaggle dataset, achieving model metrics of 88% precision, 87% recall, and 87.5% F1-score.

- Designed scalable modular architecture with separate components for data processing, model logic, and web interface to support future enhancements like personality assessments and mobile accessibility.

NE555 PWM Motor Speed Controller | *Proteus, Circuit Design, PWM*

GitHub: [Link](#)

- Designed and simulated a Pulse Width Modulation (PWM) based DC motor speed controller using NE555 timer IC in Proteus simulation software.
- Implemented variable duty cycle control circuit to regulate motor speed efficiently, demonstrating practical application of analog electronics and timing circuits.
- Analyzed PWM signal characteristics and motor response to validate circuit performance, ensuring precise speed control across different operating conditions.
- Created comprehensive circuit documentation including component specifications, waveform analysis, and performance optimization techniques.

Real-Time Parking Slot Monitoring System | *8051 Microcontroller, Embedded C, Proteus*

GitHub: [Link](#)

- Designed and implemented an automated parking management system using 8051 microcontroller to monitor real-time parking slot availability and control entry/exit gates.
- Programmed embedded C firmware to handle vehicle detection via SPST sensors, update parking count dynamically, and control DC motors through L293D motor driver for automated gate operation.
- Interfaced 16x2 LCD module to display real-time parking availability and system status messages, providing clear user feedback and reducing manual supervision requirements.
- Implemented robust debouncing logic and error handling mechanisms to prevent false triggering, achieving stable performance with zero counting errors during simulation.
- Validated complete hardware design and software logic through Proteus simulation, demonstrating reliable operation for deployment in small to medium-scale parking facilities.

BinBuddy: Powered by Arduino | *Arduino, IoT, C++*

GitHub: [Link](#)

- Designed and developed an IoT-based smart dustbin system using Arduino Uno, HC-SR04 ultrasonic sensor, and SG90 servo motor for automated, contactless waste disposal.
- Implemented real-time distance sensing to detect hand proximity and trigger automatic lid opening, enhancing hygiene and convenience in various environments.
- Programmed and integrated sensor-actuator logic for responsive lid control with an average reaction time of 180 milliseconds, ensuring smooth and reliable operation.
- Focused on creating a low-cost, scalable solution suitable for educational and practical applications, with potential for future upgrades like Wi-Fi connectivity and mobile integration.

Game Manager Using C | *C, File Handling*

GitHub: [Link](#)

- Developed a console-based Game Manager System in C, featuring a menu-driven interface for enhanced user interaction.
- Implemented the Rock Paper Scissors game with an AI opponent using random number generation for dynamic gameplay.
- Applied modular programming techniques to organize code into reusable functions, ensuring maintainability and clarity.
- Handled input validation and error management to provide a smooth and robust gaming experience.
- Enabled persistent storage of game history with detailed results, showcasing usage of file handling in C.

Certifications & Achievements

Bharatiya Antariksh Hackathon 2025

- National-level hackathon on space technology — collaborated in a multidisciplinary team; applied Python and problem-solving under time constraints to develop innovative space-tech solutions.

National Space Hackathon 2025

IIT Delhi

- Participated in prestigious national-level space technology hackathon, demonstrating expertise in space applications and innovative problem-solving approaches.

Quantathon - Shaastra 2026

IIT Madras

- Selected for the final round of Quantathon 2026 at IIT Madras (Shaastra); secured 22nd rank in this multi-stage competition centered on quantitative analysis, logical reasoning, and algorithmic thinking.

Internship Program in IoT

1stop

- Completed comprehensive IoT internship program covering sensor integration, microcontroller programming, and practical implementation of Internet of Things solutions.

Getting Started with Artificial Intelligence

IBM SkillsBuild

- Gained foundational knowledge in AI concepts, machine learning fundamentals, and practical applications through IBM's professional certification program.

The Startup Ecosystem Survey Certificate

Wadhwani Foundation, KaroStartup & JITSIE IIT Madras

- Participated in a comprehensive survey on startup ecosystem dynamics, gaining insights into entrepreneurial trends, challenges, and opportunities in the Indian startup landscape.

The Venture Capital**E-Cell, IIT (BHU)**

- Completed certification on venture capital fundamentals, startup funding mechanisms, and investment evaluation strategies from IIT BHU's Entrepreneurship Cell.

Product Management**Nextleap & Cognizance 2025, IIT Roorkee**

- Acquired comprehensive knowledge in product lifecycle management, user research, roadmap planning, and agile methodologies through IIT Roorkee's technical fest program.

Cybersecurity and Ethical Hacking**RV Institute of Technology and Management**

- Completed hands-on certification covering network security, penetration testing, vulnerability assessment, and ethical hacking techniques for securing digital systems.

Astrophysics and Cosmology**Ethical Edufabrica Pvt. Ltd. & Pravega 2024, IISc Bangalore**

- Explored advanced concepts in astrophysics, cosmology, and space science through IISc Bangalore's prestigious science fest certification program.